

Review Questions

The answers to the chapter review questions can be found in the Appendix.

- Which statements about the `final` modifier are correct? (Choose all that apply.)
 - A. Instance and static variables can be marked `final`. las instancias y static pueden ser final, en el caso E un primitivo no puede recibir otro valor despues de que se inicializo
 - B. A variable is effectively final only if it is marked `final`.
 - C. An object that is marked `final` cannot be modified.
 - D. Local variables cannot be declared with type `var` and the `final` modifier.
 - E. A primitive that is marked `final` cannot be modified.
- Which of the following can fill in the blank in this code to make it compile? (Choose all that apply.)

```
public class Ant {
    ____ void method() {}
}
```

B,C son los modificadores validos para que el codigo compile

- A. `default`
 - B. `final`
 - C. `private`
 - D. `Public`
 - E. `String`
 - F. `zzz:`
- Which of the following methods compile? (Choose all that apply.)
 - A. `final static void rain() {}`
 - B. `public final int void snow() {}`
 - C. `private void int hail() {}`
 - D. `static final void sleet() {}`
 - E. `void final ice() {}`
 - F. `void public slush() {}`

A,D ya que no importa el orden en que se usa final y static y en el segundo caso deben ir antes de void para que compile
 - Which of the following can fill in the blank and allow the code to compile? (Choose all that apply.)


```
final _____ song = 6;
```

 - A. `int`
 - B. `Integer`
 - C. `long`
 - D. `Long`
 - E. `double`
 - F. `Double`

A,B,C,E son las formas correctas en que podemos asignar la variable song en java, ademas un int en java puede convertirse a Integer mediante autoboxing y promoverse a long y double.

5. Which of the following methods compile? (Choose all that apply.)

- A. `public void january() { return; }`
- B. `public int february() { return null; }`
- C. `public void march() { }`
- D. `public int april() { return 9; }`
- E. `public int may() { return 9.0; }`
- F. `public int june() { return; }`

A,C,D son las formas correctas para declarar de tal forma que devolvamos o no un valor

6. Which of the following methods compile? (Choose all that apply.)

- A. `public void violin(int... nums) { }`
- B. `public void viola(String values, int... nums) { }`
- C. `public void cello(int... nums, String values) { }`
- D. `public void bass(String... values, int... nums) { }`
- E. `public void flute(String[] values, ...int nums) { }`
- F. `public void oboe(String[] values, int[] nums) { }`

aqui tenemos que podemos tener parametros antes del varargs, ademas son arreglos normales por lo que no existen restricciones de varargs.

7. Given the following method, which of the method calls return 2? (Choose all that apply.)

```
public int juggle(boolean b, boolean... b2) {
    return b2.length;
}
```

- A. `juggle();`
- B. `juggle(true);`
- C. `juggle(true, true);`
- D. `juggle(true, true, true);`
- E. `juggle(true, {true, true});`
- F. `juggle(true, new boolean[2]);`

el primer boolean apunta a b mientras que los otros forman un varargs de tamaño 2 y en F se pasa de forma directa un arreglo de longitud 2.

8. Which of the following statements is correct?

- A. Package access is more lenient than protected access.
- B. A public class that has private fields and package methods is not visible to classes outside the package.
- C. You can use access modifiers so only some of the classes in a package see a particular package class.
- D. You can use access modifiers to allow access to all methods and not any instance variables.
- E. You can use access modifiers to restrict access to all classes that begin with the word Test.

D es la unica descripcion correcta

9. Given the following class definitions, which lines in the `main()` method generate a compiler error? (Choose all that apply.)

// Classroom.java

```
package my.school;

public class Classroom {
    private int roomNumber;
    protected static String teacherName;
    static int globalKey = 54321;
    public static int floor = 3;
    Classroom(int r, String t) {
        roomNumber = r;
        teacherName = t; } }
```

// School.java

```
1: package my.city;
2: import my.school.*;
3: public class School {
4:     public static void main(String[] args) {
5:         System.out.println(Classroom.globalKey);
6:         Classroom room = new Classroom(101, "Mrs. Anderson");
7:         System.out.println(room.roomNumber);
8:         System.out.println(Classroom.floor);
9:         System.out.println(Classroom.teacherName); } }
```

A. None: the code compiles fine.

B. Line 5

C. Line 6

D. Line 7

E. Line 8

F. Line 9

en este caso `globalKey` tiene acceso de paquete mientras que `School` se encuentra en otro, despues el constructor tiene acceso de paquete pero no es visible desde `my.city`, `roomNumber` es privado y solo lo podemos usar dentro de `Classroom`, `teacherName` es `protected`.

10. What is the output of executing the Chimp program?

// Rope.java

```
1: package rope;
2: public class Rope {
3:     public static int LENGTH = 5;
4:     static {
5:         LENGTH = 10;
6:     }
7:     public static void swing() {
```

```

8:      System.out.print("swing ");
9:  } }

```

// Chimp.java

```

1: import rope.*;
2: import static rope.Rope.*;
3: public class Chimp {
4:     public static void main(String[] args) {
5:         Rope.swing();
6:         new Rope().swing();
7:         System.out.println(LENGTH);
8:     } }

```

El inicializador static cambia el tamaño de 5 a 10 antes de que se use la clase

- A. swing swing 5
- B. swing swing 10**
- C. Compiler error on line 2 of Chimp
- D. Compiler error on line 5 of Chimp
- E. Compiler error on line 6 of Chimp
- F. Compiler error on line 7 of Chimp

11. Which statements are true of the following code? (Choose all that apply.)

```

1: public class Rope {
2:     public static void swing() {
3:         System.out.print("swing");
4:     }
5:     public void climb() {
6:         System.out.println("climb");
7:     }
8:     public static void play() {
9:         swing();
10:        climb();
11:    }
12:    public static void main(String[] args) {
13:        Rope rope = new Rope();
14:        rope.play();
15:        Rope rope2 = null;
16:        System.out.print("-");
17:        rope2.play();
18:    } }

```

- A. The code compiles as is.
- B. There is exactly one compiler error in the code. en la linea 10 se intenta llamar a climb() que es de instancia dentro de un método static
- C. There are exactly two compiler errors in the code.
- D. If the line(s) with compiler errors are removed, the output is swing-climb.
- E. If the line(s) with compiler errors are removed, the output is swing-swing.
- F. If the line(s) with compile errors are removed, the code throws a NullPointerException.

12. How many variables in the following method are effectively final?

```

10: public void feed() {
11:     int monkey = 0;
12:     if(monkey > 0) {
13:         var giraffe = monkey++;
14:         String name;
15:         name = "geoffrey";
16:     }
17:     String name = "milly";
18:     var food = 10;
19:     while(monkey <= 10) {
20:         food = 0;
21:     }
22:     name = null;
23: }

```

- A. 1
- B. 2 aqui se tiene que solo giraffe y el primer nombre son effectively final
- C. 3
- D. 4
- E. 5
- F. None of the above. The code does not compile.

13. What is the output of the following code?

```

// RopeSwing.java
import rope.*;
import static rope.Rope.*;
public class RopeSwing {
    private static Rope rope1 = new Rope();
    private static Rope rope2 = new Rope();
    {
        System.out.println(rope1.length);
    }
}

```

```
public static void main(String[] args) {  
    rope1.length = 2;  
    rope2.length = 8;  
    System.out.println(rope1.length);  
}  
}
```

// Rope.java

```
package rope;  
public class Rope {  
    public static int length = 0;  
}
```

- A.** 02
B. 08
C. 2
D. 8
E. The code does not compile.
F. An exception is thrown.
- tenemos que lenght es statis por lo que la ultima asignacion deja el valor en 8

14. How many lines in the following code have compiler errors?

```
1: public class RopeSwing {  
2:     private static final String leftRope;  
3:     private static final String rightRope;  
4:     private static final String bench;  
5:     private static final String name = "name";  
6:     static {  
7:         leftRope = "left";  
8:         rightRope = "right";  
9:     }  
10:    static {  
11:        name = "name";  
12:        rightRope = "right";  
13:    }  
14:    public static void main(String[] args) {  
15:        bench = "bench";  
16:    }  
17: }
```

- A.** 0
B. 1

si una variable es static final, debe ser declarada solo 1 vez

- C. 2
- D. 3
- E. 4**
- F. 5

15. Which of the following can replace line 2 to make this code compile? (Choose all that apply.)

```
1: import java.util.*;
2: // INSERT CODE HERE
3: public class Imports {
4:     public void method(ArrayList<String> list) {
5:         sort(list);
6:     }
7: }
```

B este bloque import static con el comodin
importa metodos static como sort()

- A. import static java.util.Collections;
- B. import static java.util.Collections.*;**
- C. import static java.util.Collections.sort(ArrayList<String>);
- D. static import java.util.Collections;
- E. static import java.util.Collections.*;
- F. static import java.util.Collections.sort(ArrayList<String>);

16. What is the result of the following statements?

```
1: public class Test {
2:     public void print(byte x) {
3:         System.out.print("byte-");
4:     }
5:     public void print(int x) {
6:         System.out.print("int-");
7:     }
8:     public void print(float x) {
9:         System.out.print("float-");
10:    }
11:    public void print(Object x) {
12:        System.out.print("Object-");
13:    }
14:    public static void main(String[] args) {
15:        Test t = new Test();
16:        short s = 123;
17:        t.print(s);
18:        t.print(true);
19:    }
20: }
```

```

19:      t.print(6.789);
20:  }
21: }

```

- A. byte-float-Object-
- B. int-float-Object-
- C. byte-Object-float-
- D. int-Object-float-
- E. int-Object-Object-
- F. byte-Object-Object-

aqui tenemos que short lo promovemos a int mientras que double y boolean se autoboxean

17. What is the result of the following program?

```

1: public class Squares {
2:     public static long square(int x) {
3:         var y = x * (long) x;
4:         x = -1;
5:         return y;
6:     }
7:     public static void main(String[] args) {
8:         var value = 9;
9:         var result = square(value);
10:        System.out.println(value);
11:    } }

```

- A. -1
- B. 9
- C. 81
- D. Compiler error on line 9
- E. Compiler error on a different line

aqui pasamos argumentos por valor, por lo que cambiar x no modifica su valor

18. Which of the following are output by the following code? (Choose all that apply.)

```

public class StringBuilders {
    public static StringBuilder work(StringBuilder a,
        StringBuilder b) {
        a = new StringBuilder("a");
        b.append("b");
        return a;
    }
    public static void main(String[] args) {
        var s1 = new StringBuilder("s1");
        var s2 = new StringBuilder("s2");

```

```

    var s3 = work(s1, s2);
    System.out.println("s1 = " + s1);
    System.out.println("s2 = " + s2);
    System.out.println("s3 = " + s3);
}
}

```

- A. `s1 = a`
- B. `s1 = s1` primero tenemos que si reasignamos a dentro del metodo no cambia su referencia, luego append() modifica el StringBuilder al que apunta s2
- C. `s2 = s2`
- D. `s2 = s2b`
- E. `s3 = a`
- F. The code does not compile.

19. Which of the following will compile when independently inserted in the following code? (Choose all that apply.)

```

1: public class Order3 {
2:     final String value1 = "red";
3:     static String value2 = "blue";
4:     String value3 = "yellow";
5:     {
6:         // CODE SNIPPET 1
7:     }
8:     static {
9:         // CODE SNIPPET 2
10:    } }

```

- A. Insert at line 6: `value1 = "green";`
- B. Insert at line 6: `value2 = "purple";` un inicializador de instancia puede modificar una variable static no final, asi como un campo de instancia, y un inicializador static puede modificar una variable static no final
- C. Insert at line 6: `value3 = "orange";`
- D. Insert at line 9: `value1 = "magenta";`
- E. Insert at line 9: `value2 = "cyan";`
- F. Insert at line 9: `value3 = "turquoise";`

20. Which of the following are true about the following code? (Choose all that apply.)

```

public class Run {
    static void execute() {
        System.out.print("1-");
    }
    static void execute(int num) {
        System.out.print("2-");
    }
}

```

```

static void execute(Integer num) {
    System.out.print("3-");
}
static void execute(Object num) {
    System.out.print("4-");
}
static void execute(int... nums) {
    System.out.print("5-");
}
public static void main(String[] args) {
    Run.execute(100);
    Run.execute(100L);
}
}

```

- A.** The code prints out 2-4-. aquí tenemos que 100 es int y 100L autoboxea Long y usa Object
- B.** The code prints out 3-4-.
- C.** The code prints out 4-2-.
- D.** The code prints out 4-4-.
- E.** The code prints 3-4- if you remove the method `static void execute(int num)`.
- F.** The code prints 4-4- if you remove the method `static void execute(int num)`.

21. Which method signatures are valid overloads of the following method signature? (Choose all that apply.)

```
public void moo(int m, int... n)
```

- A.** `public void moo(int a, int... b)`
- B.** `public int moo(char ch)` se cambia la lista de parametros por una sobrecarga valida y en
- C.** `public void moooo(int... z)` D int equivale a int[] y su firma no coincide con (int,int[]).
- D.** `private void moo(int... x)`
- E.** `public void moooo(int y)`
- F.** `public void moo(int... c, int d)`
- G.** `public void moo(int... i, int j...)`